

Shanu Kumar Manglam

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PROFESSIONAL SUMMARY

Results-driven Software Engineer specializing in **Generative AI, Machine Learning, NLP, LLMs, Retrieval-Augmented Generation (RAG), Data Engineering, ETL Pipelines, and Intelligent Automation**. Experienced in designing scalable AI applications using Python, LangChain, FAISS, Streamlit, SQL, REST APIs, and cloud technologies. Skilled in semantic search, prompt engineering, vector databases, workflow automation, Agile development, and CI/CD with a strong focus on performance optimization and business impact.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL

Artificial Intelligence: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Prompt Engineering, Context Engineering

RAG & Vector Databases: LangChain, FAISS, Semantic Search, Embedding Models, Retrieval-Augmented Generation

Frameworks & Tools: Streamlit, GitHub Actions, Docker, Playwright, REST APIs, Power BI, Looker Studio

Data Engineering: ETL Pipelines, Apache Spark (PySpark), Data Warehousing, Data Transformation, Data Quality

Cloud & DevOps: AWS, Git, GitHub, GitHub Copilot, Kubernetes (Basics), CI/CD

Analytics: Google Analytics 4 (GA4), Google Tag Manager, Adobe Analytics

Methodologies: Agile, Scrum

PROFESSIONAL EXPERIENCE

Cognizant - Programmer Analyst Trainee | Chennai, India

Feb 2026 – Apr 2026

- Automated digital marketing workflows using Microsoft Copilot Studio and GitHub Copilot, reducing manual effort by approximately **30%**.
- Engineered scalable ETL pipelines using Python and PySpark for processing large marketing datasets, improving reporting efficiency by **25%**.
- Designed interactive dashboards integrating GA4, GTM, Adobe Analytics, and Looker Studio for campaign performance monitoring.
- Implemented LLM-based automation and prompt engineering workflows for intelligent content generation, document summarization, and reporting.
- Collaborated within Agile teams using Git-based version control while following SDLC best practices.

Ernst & Young - Data Analyst Intern | Remote

Feb 2024 – Apr 2024

- Developed predictive analytics models and business intelligence dashboards using Python, SQL, and Power BI.
- Performed ETL, data cleansing, and transformation on structured datasets, improving reporting accuracy by approximately **20%**.
- Created KPI dashboards enabling data-driven decision-making across healthcare analytics projects.
- Presented analytical insights using data visualization and statistical analysis techniques.

PROJECTS

HR Policy Assistant – RAG Application | FAISS, LangChain, GPT-4o, Docker

May 2026

- Engineered an enterprise Retrieval-Augmented Generation (RAG) assistant using LangChain, GPT-4o, FAISS, and Streamlit for intelligent HR policy question answering.
- Implemented semantic search with vector embeddings, improving document retrieval relevance by approximately **35%**.
- Integrated REST APIs and prompt engineering techniques to generate context-aware responses.
- Containerized deployment using Docker and automated CI/CD workflows with GitHub Actions.

ATS Resume Copilot – Agentic Job Search Automation | Gemini API, Playwright, Streamlit

Apr 2026

- Designed an AI-powered job application automation platform integrating Gemini API, Playwright, LangChain, and Streamlit.
- Automated resume analysis, ATS optimization, job scraping and application tracking, reducing manual effort by over **50%**.
- Implemented workflow automation, authentication handling, and browser automation for scalable job applications.

Research Paper Retrieval System | Transformers, FAISS, Streamlit

Mar 2026

- Built an intelligent academic search engine using Retrieval-Augmented Generation (RAG), FAISS vector databases, and embedding models.
- Optimized semantic retrieval pipelines, improving search accuracy and response quality for research documents.
- Implemented chunking, vector indexing, metadata filtering, and document ranking for scalable retrieval.

EDUCATION

University of Engineering & Management, Kolkata Bachelor of Technology in Electronics and Communication Engineering	Aug 2021 – Jul 2025 CGPA: 8.86 / 10
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AWARDS

Vice Chancellor’s Award for Best Scientific Mind 2025 	May 2025
• Awarded for outstanding innovation and research contributions in Artificial Intelligence, Machine Learning , and Electronics Engineering .	

PUBLICATIONS

Review Paper: Renewable Energy Trends for the Next Decade <i>UEMGREEN’25</i> 	Apr 2025
• Published a review on AI-driven renewable energy technologies, grid modernization, sustainable energy systems, and policy-driven clean energy adoption.	
Research Paper: Analysing CO₂ Emission from Vehicles Using Machine Learning <i>UEMGREEN’25</i> 	Apr 2025
• Developed and evaluated Machine Learning models for predicting vehicular CO ₂ emissions, supporting sustainability analytics and climate action (SDG 13).	

CERTIFICATIONS

Generative AI - Generative AI for Beginners - Prompt Engineering Foundation - Context Engineering Foundation - Responsible AI (Cognizant)
Developer Tools - Microsoft Copilot Studio - GitHub Copilot Complete Guide - Apache Spark 3 – Data Engineering
Cloud & Analytics - AWS Cloud Fundamentals - Google Analytics 4 (GA4) - Google Tag Manager (GTM) - IBM Data Science Professional Certificate